NV-Center Optical Readout of Superconducting Bolometer Pixels for Ultra-Low Threshold Particle Detection

1st Jacob Bogenschuetz
Department of Physics
University of Texas at Arlington
Arlington, USA
jacob.bogenschuetz@uta.edu

2nd Brad Brown
Department of Physics
University of Texas at Arlington
Arlington, USA
bab0345@mavs.uta.edu

3rd V. A. Chirayath
Department of Physics
University of Texas at Arlington
Arlington, USA
chirayat@uta.edu

4th Jonathan Asaadi
Department of Physics
University of Texas at Arlington
Arlington, USA
jonathan.asaadi@uta.edu

5th B. J. P. Jones
Department of Physics
University of Texas at Arlington
Arlington, USA
ben.jones@uta.edu

***Abstract*—We present a bolometric detector architecture that leverages nitrogen-vacancy (NV) center magnetometry to probe superconducting thin-film pixels, and report extensive cryogenic characterization of the NV-diamond sensing element from 11 K to 200 K. The detector concept exploits the thermal sensitivity of the Meissner effect: energy deposition from a particle heats a superconducting pixel past its critical temperature, collapsing Meissner screening and allowing an applied magnetic field to reach nearby NV centers. The resulting shift in optically detected magnetic resonance (ODMR) frequency produces a fluorescence plateau whose duration encodes the deposited energy. We describe the device architecture and present simulation results for a proof-of-principle NbN/Ge demonstrator targeting MeV-scale energy deposition. We report measurements of the zero-field splitting $D(T)$, transverse strain parameter E, ODMR contrast, and spin-lattice relaxation time T_1 as functions of temperature, establishing the operating parameters required for detector design. Additionally, we demonstrate the platform's multi-modal sensing capability through real-time broadband Raman monitoring of structural phase transitions in amorphous solid water, observed serendipitously during cryostat operation. These results validate the cryogenic NV-diamond platform and outline the pathway to sub-eV threshold detection relevant for light dark matter searches.**

***Index Terms*—quantum sensing, NV centers, superconducting detectors, ODMR, bolometer, magnetometry, cryogenic diamond, amorphous ice**

I. Introduction

Superconducting bolometers offer exceptional energy resolution for particle detection, but their readout—typically via transition-edge sensors (TES) or kinetic inductance detectors (KIDs)—requires complex cryogenic multiplexing using SQUID amplifiers and microwave feedlines [1]. This wiring overhead limits scalability and introduces thermal loads that constrain detector performance.

We propose a fundamentally different approach: using the optical magnetometry capabilities of nitrogen-vacancy (NV) centers in diamond to read out superconducting pixels with no electrical connection to the cold stage. NV centers provide shot-noise-limited magnetic field sensitivity across a wide dynamic range [2], [3], and their optical interface enables straightforward multiplexing via imaging. This architecture offers a pathway to large-area, densely pixelated detectors with sub-eV energy thresholds.

The operating principle exploits the temperature dependence of the London penetration depth $\lambda(T)$ in superconductors. Below T_c, Meissner screening excludes an applied magnetic field from the bulk. Energy deposition increases the local temperature, increasing $\lambda(T)$ and allowing field penetration. NV centers positioned near the superconducting surface detect this field change through shifts in their spin resonance frequencies.

Realizing this concept requires detailed understanding of NV-center behavior at cryogenic temperatures. In this work, we present comprehensive characterization of an ensemble NV-diamond sensor from 11 K to 200 K, including the temperature dependence of the zero-field splitting $D(T)$, transverse strain parameter E, ODMR contrast, and spin-lattice relaxation time T_1. These measurements establish the operating envelope for detector design.

II. Device Architecture

A. Layer Stack and Geometry

The detector stack comprises (from top): a sapphire substrate with transparent ITO electrode, a shallow NV-diamond layer (100 nm depth), a 40 nm Au film serving as both optical filter and thermal strap, a 100 nm SiO_2 thermal isolation layer, and lithographically defined superconducting pixels with optional absorber material.

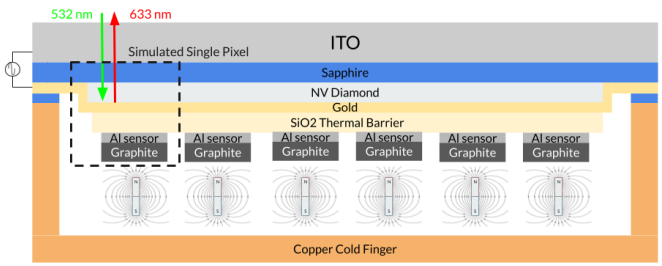

Fig. 1. Conceptual device stack and optical geometry showing an array of superconducting Al/graphite pixels with individual bias magnets on the copper cold finger. The dashed box indicates a single simulated pixel. Microwave drive for ODMR is capacitively coupled between the ITO and Au electrodes.

The NV ensemble is excited at 532 nm, and fluorescence near 633 nm is collected in an epifluorescence geometry through the same objective. A microwave field drives spin transitions for ODMR. The Au layer blocks residual pump light from reaching the superconductor while providing lateral thermal contact to the cold finger.

B. Demonstrator Configuration

For proof-of-principle validation, we target a demonstrator pixel using 200 nm NbN ($T_c \simeq 15.7\,\mathrm{K}$) with a $1\,\mu$m Ge absorber on a $1.5 \times 5\,\mu\mathrm{m}^2$ footprint. The narrow width mitigates vortex nucleation under the $\sim$1 mT bias field required for ODMR operation [5]. Operating near $T_{\mathrm{op}} \approx 12\,\mathrm{K}$ provides thermal headroom while maintaining strong Meissner screening.

This configuration targets validation with 5.5 MeV alpha particles from ^{241}Am, producing plateau durations $\tau \sim \mathcal{O}(10^2)\,\mu$s resolvable with standard photodetection electronics. Cross-talk between adjacent pixels is mitigated by the $1/r^3$ fall-off of the magnetic dipole field from each pixel and can be further suppressed through spatial deconvolution of the wide-field fluorescence image. The pixel pitch is chosen so that the field from a neighboring pixel at the NV layer is below the ODMR linewidth.

III. Readout Principle and Simulation

A. ODMR-Based Transduction

The NV ground-state spin Hamiltonian in an external magnetic field is

$$\mathcal{H} = D\,S_z^2 + E(S_x^2 - S_y^2) + \gamma_e \vec{B} \cdot \vec{S} \tag{1}$$

where $D \simeq 2877\,\mathrm{MHz}$ is the zero-field splitting at cryogenic temperatures, E is the transverse strain parameter, and $\gamma_e \simeq 2.8\,\mathrm{MHz/G}$ is the electron gyromagnetic ratio [3], [6]. Under continuous-wave ODMR with a fixed microwave drive frequency ν_{MW}, the fluorescence exhibits a Lorentzian dip when ν_{MW} matches a spin resonance.

We operate at $T_{\mathrm{op}} < T_c$ with ν_{MW} tuned to the ODMR dip minimum. When energy deposition raises $T > T_c$, Meissner screening collapses, the local field at the NV layer increases, and the resonance detunes from ν_{MW}. The fluorescence rises from the ODMR dip to baseline, forming a plateau whose duration τ encodes the deposited energy.

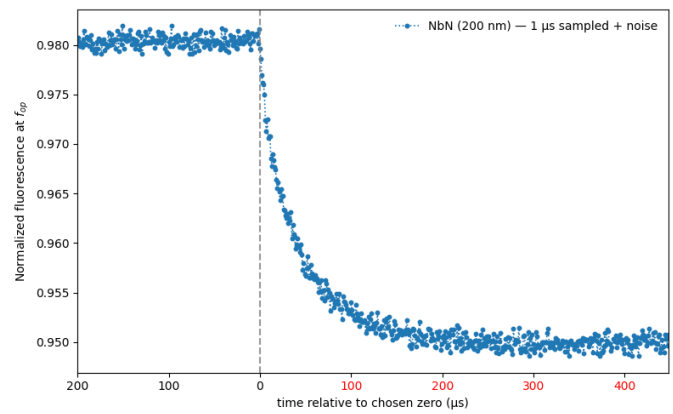

Fig. 2. Simulated NbN/Ge readout trace at $1\,\mu$s sampling with Gaussian fluorescence noise and $K_\uparrow = 3/K_\downarrow = 2$ trigger. The plateau remains clearly resolvable.

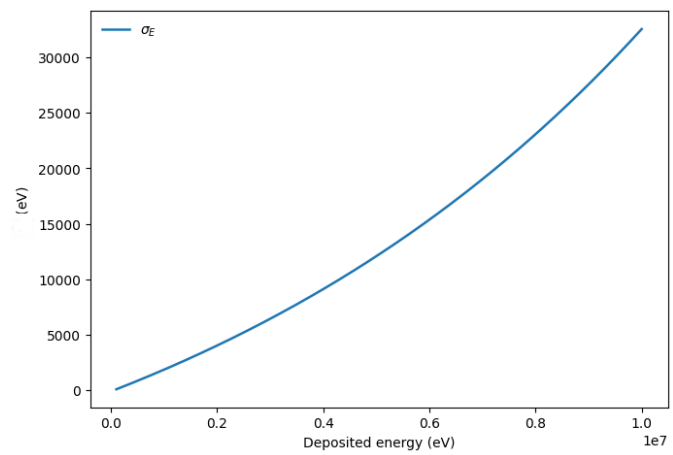

Fig. 3. Energy resolution for the NbN/Ge demonstrator at $T_{\mathrm{op}} \approx 12\,\mathrm{K}$. Sub-percent fractional resolution is predicted across the MeV region of interest.

B. Energy Reconstruction via Timing

The transduction chain connects thermal response, London penetration depth, magnetic field at the NV center, and fluorescence intensity:

$$E_{\mathrm{dep}} \to T(t) \to \lambda(T) \to B_{\mathrm{NV}}(t) \to F(t) \to \tau(E_{\mathrm{dep}}) \tag{2}$$

where the London depth follows the Gorter–Casimir form $\lambda(T) = \lambda_0/\sqrt{1-(T/T_c)^4}$ [7].

Monte Carlo simulations including fluorescence intensity noise ($\sigma_F = 7.06 \times 10^{-4}$) [5], $1\,\mu$s sampling quantization, and random phase offsets demonstrate that plateau edges are resolvable with a trigger requiring $K = 3$ consecutive threshold crossings (Fig. 2). The energy resolution follows $\sigma_E \simeq |dE/d\tau|\,\sigma_\tau$, where σ_τ is the timing jitter (Fig. 3).

For the NbN/Ge demonstrator, simulations predict sub-percent fractional energy resolution ($\sigma_E/E < 1\%$) across the

MeV region of interest, with plateau durations of hundreds of microseconds.

IV. Cryogenic NV-Diamond Characterization

To establish the operational envelope for our NV-diamond sensor, we characterize its cryogenic performance from 11 K to 200 K—demonstrating techniques and stability that directly translate to the final detector design. Measurements are performed on a commercial HPHT diamond (Adamas Nano, [NV] ∼2 ppm) with an ensemble NV layer, cooled in a closed-cycle cryostat (Lakeshore 335 temperature controller). ODMR spectra are acquired using a 532 nm diode laser, ADF4350 microwave synthesizer, and Thorlabs PM101 power meter.

A. Zero-Field Splitting $D(T)$ and Strain E

At zero applied magnetic field, the NV ground-state Hamiltonian (1) yields two ODMR transitions at frequencies $f_\pm = D \pm |E|$, where $|E| = \sqrt{E_x^2 + E_y^2}$ is the magnitude of the transverse strain [3]. We performed a temperature sweep from 11 K to 200 K with the permanent magnet retracted, recording ODMR spectra at each temperature stop.

Fig. 4 shows the two zero-field ODMR dip frequencies and their midpoint $D(T) = (f_+ + f_-)/2$. The zero-field splitting is approximately constant at $D \approx 2877.5$ MHz below 50 K, then decreases monotonically to $D \approx 2875.3$ MHz at 200 K. This behavior is well described by the two-phonon Bose–Einstein model of Cambria *et al.* [10]:

$$D(T) = D_0 + \sum_{i=1}^{2} \frac{c_i}{e^{\Delta_i/k_B T} - 1} \tag{3}$$

where $D_0 \approx 2877.7$ MHz is the zero-temperature value and the two phonon modes ($\Delta_1 \approx 59$ meV, $\Delta_2 \approx 146$ meV) account for the dominant electron-phonon (∼85%) and thermal expansion (∼15%) contributions [9]. At our operating temperature of 12 K, both phonon modes are exponentially frozen ($n(\Delta_1/k_B \cdot 12\,\mathrm{K}) \sim e^{-57} \approx 0$), making D thermally stable to better than 10 kHz per 0.1 K—far below our 0.1 MHz bin resolution.

The transverse strain parameter E is extracted from $2E = f_+ - f_-$ at zero field. At cryogenic temperatures we measure $E \approx 1.2$ MHz, consistent with typical ensemble HPHT diamond values [3]. Although apparent variations in E can arise when superimposed peaks from inequivalent NV sub-ensembles shift their relative contrast with temperature, the intrinsic transverse crystal-field splitting is temperature-independent [3]. We therefore treat E as a fixed material parameter in our Hamiltonian model.

B. ODMR Contrast versus Temperature

The ODMR contrast—defined as the relative fluorescence dip depth—determines the signal amplitude available for plateau detection. We measured contrast versus temperature at $B \approx 43$ G (per-axis dip) and at zero field (total dip).

Fig. 5 overlays both datasets. Both exhibit a striking non-monotonic profile. According to Ernst *et al.* [12] and Happacher *et al.* [13], who independently developed quantitative models of this behavior, the temperature dependence arises from the interplay of spin mixing and phonon-mediated orbital hopping in the NV ^3E excited state (ES). The ES possesses two orbital branches, E_x and E_y, split by crystal strain $\delta_\perp$, each carrying three spin sublevels ($m_s = 0, \pm 1$). Three distinct regimes emerge:

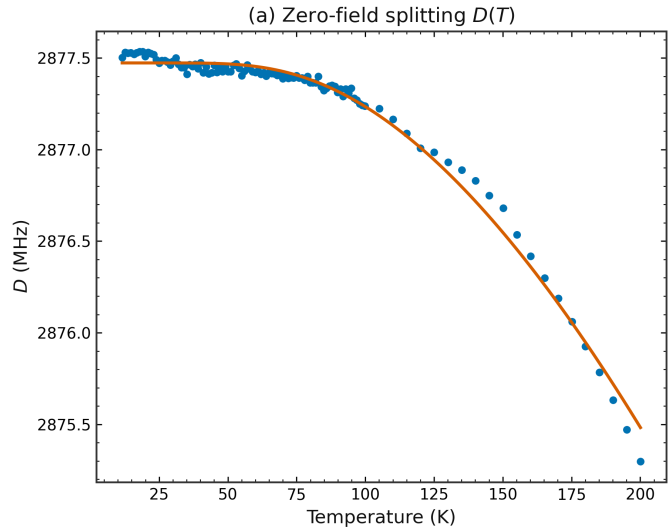

Fig. 4. Zero-field ODMR dip frequencies $f_\pm = D \pm E$ and extracted $D(T)$ versus temperature from 11 K to 200 K, with the two-phonon Bose–Einstein fit of Eq. (3).

(I) $T \lesssim 10$ **K:** Orbital hopping between E_x and E_y is slower than the ES radiative decay rate ($\tau_E^{-1} \approx 10^8\,\mathrm{s}^{-1}$), so spin states are largely preserved through the optical cycle. The spin-selective intersystem crossing (ISC) via the $^1A_1$ singlet—which preferentially shelves $m_s = \pm 1$ population [14]—operates efficiently, yielding high contrast (∼5–7%).

(II) ∼10–100 K: Two-phonon Raman orbital hopping ($\Gamma_{XY} \propto T^5$) accelerates until it approaches the Larmor precession frequency set by the ES spin-orbit splittings [12]. Frequent orbital jumps randomize the spin-phase evolution between $|0\rangle$ and $|\pm 1\rangle$ across the two branches, collapsing the spin polarization that underpins ODMR contrast. This resonance condition—maximally destructive when the hopping period matches approximately half the Larmor precession period—produces the observed contrast minimum near 30–40 K.

(III) $T \gtrsim 100$ **K:** Hopping far exceeds the spin-mixing rate, entering a motional-narrowing regime. The orbital branches time-average into an effective singlet resembling the $^3A_2$ ground state, restoring $|0\rangle$ and $|\pm 1\rangle$ as pure spin eigenstates and recovering spin-selective ISC [13]. Contrast peaks near 75 K before thermally activated depolarization erodes it at higher temperatures.

Crucially, the data reveal that this non-monotonic profile is itself field-dependent: the minimum depth, its temperature, and the recovery slope all shift between zero field and $B \approx 43$ G. This follows directly from the orbital-hopping framework: increasing B raises the ES Larmor frequencies, requiring faster

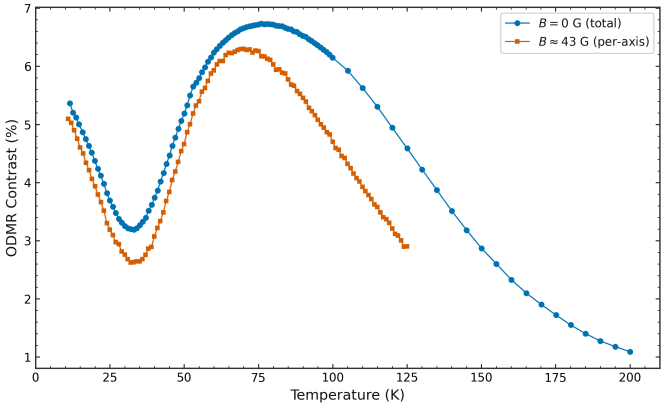

Fig. 5. ODMR contrast versus temperature at zero field (total dip contrast, blue) and $B \approx 43\,\mathrm{G}$ (per-axis dip contrast, orange). Both datasets exhibit a non-monotonic profile with a minimum near 30 K and maximum near 75 K, arising from the competition between ES orbital hopping and spin-mixing dynamics [12], [13]. The field-dependent shift of the minimum and recovery slope reflects the change in ES Larmor frequencies with applied field.

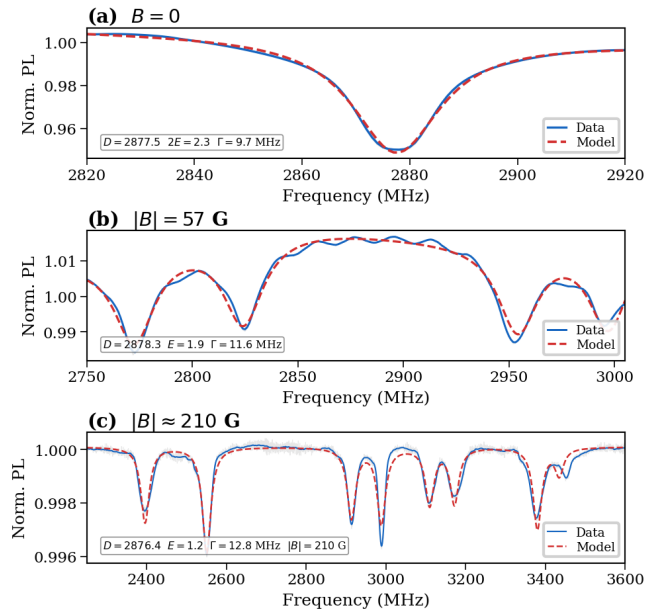

Fig. 6. ODMR spectra at $\sim$12 K with Hamiltonian fits (dashed red). Dip positions are set by the spin Hamiltonian (Eq. 5); per-dip contrasts $C_{s,i}$ are free parameters in the lineshape fit (Eq. 4). (a) Zero field: strain-split doublet at $D \approx 2877.5\,\mathrm{MHz}$, $2E \approx 2.3\,\mathrm{MHz}$. (b) $|B| \approx 57\,\mathrm{G}$: four resolved dips from two degenerate NV-axis pairs. (c) $|B| \approx 210\,\mathrm{G}$: all eight dips resolved with significant per-dip amplitude asymmetry from strain, laser polarization, and antenna geometry.

hopping (higher T) to satisfy the resonance condition, and simultaneously reduces the spin-mixing amplitudes, yielding a shallower minimum [12], [13]. That both field conditions reproduce the same qualitative three-regime structure confirms the behavior is an intrinsic property of the NV center.

The contrast minimum near 30 K defines a sensitivity dead zone to avoid in detector design. Our operating point ($T_{\mathrm{op}} \approx 12\,\mathrm{K}$) lies in Regime I with adequate contrast ($\sim$5%), though the global maximum near 75 K may motivate future designs using higher-T_c superconductors.

C. Peak Fitting and Multi-Axis Zeeman Spectroscopy

We fit the full ODMR spectrum at each field and temperature using a combined model. The Hamiltonian-derived peak positions from Eq. (5) determine the Lorentzian center frequencies, while the per-dip contrast $C_{s,i}(T)$ is treated as a free parameter extracted from the fit at each temperature. Physically, $C_{s,i}$ reflects the combined effect of the microwave transition matrix element, the spin-dependent differential intersystem crossing (ISC) rate [3], [4], the temperature-dependent excited-state orbital dynamics (Sec. IV-B) [12], [13], and experimental factors including laser polarization geometry and microwave antenna coupling that cannot be independently constrained in the present setup. The standard CW-ODMR lineshape [2], [6] is

$$P(f) = P_0 \left[1 - \sum_{i=1}^{N_{\mathrm{axes}}} \sum_{s=\pm} C_{s,i}(T)\, \mathcal{L}(f;\, f_{s,i},\, \Gamma) \right] \tag{4}$$

where $C_{s,i}(T)$ is the fractional dip depth for transition s on axis i, $\mathcal{L}$ is a Lorentzian with shared HWHM Γ, and $f_{s,i}$ are the transition frequencies from Eq. (5). Each $C_{s,i}$ is fitted independently, allowing the two transitions within a given axis pair to differ in amplitude. In practice, significant asymmetry is observed (up to a factor of three between partner dips at 210 G), arising from crystal strain inhomogeneity, laser polarization geometry, and microwave antenna coupling variations that break the idealized ISC symmetry. At zero field ($N_{\mathrm{axes}} = 1$, two transitions), the model reduces to the "W-shape" doublet. At high field ($N_{\mathrm{axes}} = 4$, eight transitions), the fit extracts the full field vector.

The diamond lattice supports four equivalent NV orientations along $\langle 111 \rangle$ crystallographic directions. An external magnetic field $\vec{B}$ projects differently onto each axis $\hat{n}_i$, producing up to eight distinct ODMR dips. In the perturbative regime ($\gamma_e |\vec{B}| \ll D$), the transition frequencies for axis i are [6]:

$$f_{\pm,i} = D + \frac{\gamma_e^2 B_{\perp,i}^2}{2D} \pm \sqrt{(\gamma_e B_{\parallel,i})^2 + E^2} \tag{5}$$

where $B_{\parallel,i} = \vec{B} \cdot \hat{n}_i$ and $B_{\perp,i} = \sqrt{|\vec{B}|^2 - B_{\parallel,i}^2}$. We performed calibrated field sweeps from 10 G to 36 G at 11 K. At $\sim$57 G, two pairs of NV axes remain pairwise degenerate, yielding four resolved dips (Fig. 6b). Breaking this degeneracy requires a field orientation that projects unequally onto all four $\langle 111 \rangle$ axes; at $\sim$210 G all eight dips become individually resolved (Fig. 6c). A global fit of (5) to all four dip pairs at high field yields the full field vector: $|\vec{B}| = 210\,\mathrm{G}$ with four distinct polar angles $\theta_i = 26.8°$, $45.0°$, $79.2°$, and $89.8°$ between $\vec{B}$ and each NV axis, with E fixed to the zero-field value of 1.2 MHz to avoid over-parameterization of the multi-axis fit. From the four measured splittings, the Cartesian field components can be reconstructed via the inverse projection $B_k = (\sqrt{3}/4) \sum_i s_i (\hat{n}_i)_k \cdot B_{\parallel,i}$, where $s_i = \pm 1$ is resolved

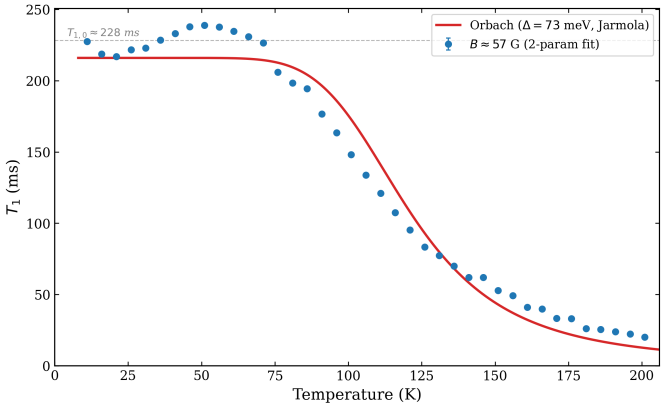

Fig. 7. Spin-lattice relaxation time T_1 versus temperature at $B \approx 57$ G. Below $\sim$65 K, T_1 plateaus at $\sim$230 ms (cross-relaxation limited). Above 65 K, the decrease is consistent with the onset of Orbach relaxation ($\Delta = 73$ meV [11]).

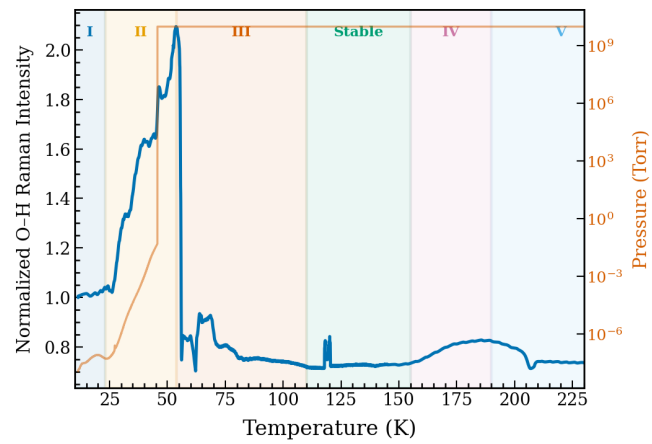

Fig. 8. Integrated O–H stretch Raman intensity (blue, left axis) and chamber pressure (orange, right axis) versus temperature during warmup. Five structural regimes are color-coded: (I) stable HDA, (II) HDA→LDA with explosive gas desorption, (III) pore collapse, (IV) crystallization, (V) sublimation.

by the known magnet geometry. This intrinsic vector magnetometry capability provides an absolute field calibration at each pixel without external reference.

D. Spin-Lattice Relaxation Time T_1

Spin-lattice relaxation (T_1) measurements were performed at $B \approx 57$ G using a three-phase pulsed protocol: baseline measurement (MW off), spin driving (MW on, 500 ms), and recovery monitoring (MW off, 1.2 s). A gold-standard 2-parameter exponential fit with the baseline pinned to the measured equilibrium value extracts T_1 from the recovery curve at each temperature (Fig. 7).

Below $\sim$65 K, $T_{1,\mathrm{off}} \approx 230$ ms remains approximately constant, indicating a cross-relaxation-limited plateau dominated by dipolar coupling to substitutional nitrogen (P1 centers) and NV–NV interactions [11]. Above 65 K, T_1 decreases smoothly, reaching $\sim$20 ms by 200 K, consistent with the onset of the Orbach two-phonon relaxation process. Jarmola *et al.* [11] report an activation energy $\Delta = 73 \pm 4$ meV for this process in NV ensembles; our data are consistent with this value but do not independently constrain Δ, as a contrast-correlated systematic in the recovery amplitude (the apparent T_1 tracks the non-monotonic ODMR contrast of Sec. IV-B) biases the extracted time constants in the intermediate regime. These long T_1 times ($\gg$ the $\sim$100 μs plateau durations expected from the demonstrator) confirm that spin relaxation does not limit the readout bandwidth.

V. Multi-Modal Sensing Capability

During a routine cooldown, the vacuum chamber was inadvertently exposed to atmospheric pressure at 11.4 K, and amorphous solid water (ASW) condensed on the cold stage. The 532 nm excitation laser probes the O–H stretch Raman band ($\sim$3200 cm^{-1} shift, emitted at $\sim$641 nm), which falls within the collection dichroic passband and was monitored continuously during warmup.

Fig. 8 shows the integrated Raman intensity and chamber pressure versus temperature. Five distinct structural regimes are resolved: (I) stable high-density amorphous ice below 23 K, (II) the HDA→LDA transition with explosive trapped-gas desorption (23–54 K), (III) pore collapse and densification (54–60 K), (IV) crystallization to hexagonal ice (155–190 K), and (V) sublimation above 200 K.

All observed transition temperatures agree with published values from infrared spectroscopy, electron diffraction, and temperature-programmed desorption studies of astrophysical ice analogues [15]. This serendipitous observation illustrates that the cryogenic optical platform can detect phase transitions in surface adsorbates as a by-product of its primary magnetometry function, without additional hardware.

VI. Roadmap and Outlook

We have defined a staged validation pathway:

D0 (Room-temperature benchtop): Demonstrate calibrated ΔB detection using wide-field NV magnetometry with lock-in ODMR readout. Gate criterion: SNR ≥ 5 at design dwell time.

D1 (Coupling NbN thin films with NV centers): Integrate NbN thin films with NV diamond at $T \approx 12$ K. Validate heater-step calibration and extract $\partial B/\partial E$. Gate: reconstruct injected energy within 20%.

D2 (Mini-array): Fabricate small NbN/Ge pixels on pre-formed NV-center heterostructures and quantify crosstalk via cross-correlation. Gate: nearest-neighbor crosstalk <5%.

D3 (Alpha validation): Expose demonstrator to ^{241}Am alphas and verify energy reconstruction against the known 5.5 MeV line.

Beyond the MeV demonstrator, the architecture scales to sub-eV detection by reducing pixel heat capacity. Simulations of Al(20 nm)/graphite(200 nm) pixels at $T_c \simeq 1.1$ K predict thresholds near 0.75 eV with $\sigma_E < 0.4$ eV—relevant for searches for light dark matter candidates including dark photons and axion-like particles [8]. The optical readout enables scaling to $\sim 10^9$ pixels via wide-field imaging of NV

fluorescence, a density impractical with electrical readout. A $(10\,\mathrm{cm})^2$ detector with $1\,\mu\mathrm{m}^2$ pixels at $2\,\mu$m pitch would contain 2.5×10^9 pixels with no electrical connections to the cold stage.

Our CW-ODMR characterization establishes a magnetic field sensitivity of $\eta \approx 14\,\mu\mathrm{T}/\sqrt{\mathrm{Hz}}$ at the drift floor, achieved with 100-pair averaging at 20 ms dwell per frequency bin. This sensitivity, combined with the measured ODMR linewidth ($\sim$10 MHz FWHM) and contrast ($>$2% at the optimal operating temperature), provides the quantitative inputs for modeling the minimum detectable field change δB during a Meissner screening event.

The cryogenic characterization presented here identifies $T_{\mathrm{op}} \approx 12\,\mathrm{K}$ as a suitable operating point, offering adequate ODMR contrast ($\sim$5%), long spin relaxation ($T_1 > 230\,\mathrm{ms}$), and well-resolved Zeeman structure. The measured $D(T)$ and E data provide the quantitative inputs needed for full Hamiltonian modeling of the NV response under varying thermal and magnetic conditions during detector operation.

VII. Conclusion

We have presented a bolometric architecture combining superconducting Meissner screening with NV-center optical magnetometry, supported by comprehensive cryogenic characterization of the diamond sensing element. Zero-field measurements from 11 K to 200 K establish $D(T)$ and the temperature-independent strain parameter $E \approx 1.2\,\mathrm{MHz}$. ODMR contrast exhibits a non-monotonic temperature dependence with a minimum near 30 K and maximum near 75 K, observed consistently across all four NV crystallographic orientations. Spin relaxation times exceed 230 ms below 65 K, confirming that T_1 does not limit readout bandwidth.

These measurements, combined with multi-axis Zeeman spectroscopy, validate the cryogenic NV-diamond platform for detector readout applications. The NbN/Ge demonstrator design, with simulated sub-percent energy resolution at MeV scales, provides a concrete pathway toward sub-eV threshold particle detection for next-generation quantum sensors.

Acknowledgments

This work was supported by the University of Texas at Arlington Department of Physics.

References

[1] K. D. Irwin and G. C. Hilton, "Transition-edge sensors," in *Cryogenic Particle Detection*, C. Enss, Ed. Berlin: Springer, 2005, pp. 63–150.

[2] J. F. Barry *et al.*, "Sensitivity optimization for NV-diamond magnetometry," *Rev. Mod. Phys.*, vol. 92, p. 015004, 2020.

[3] M. W. Doherty *et al.*, "The nitrogen-vacancy colour centre in diamond," *Phys. Rep.*, vol. 528, pp. 1–45, 2013.

[4] J.-P. Tetienne *et al.*, "Magnetic-field-dependent photodynamics of single NV defects in diamond: an application to qualitative all-optical magnetic imaging," *New J. Phys.*, vol. 14, p. 103033, 2012.

[5] R. T. Kapur *et al.*, "Flux trapping characterization for superconducting electronics using a cryogenic widefield NV-diamond microscope," *arXiv:2506.01906*, 2025.

[6] L. Rondin *et al.*, "Magnetometry with nitrogen-vacancy defects in diamond," *Rep. Prog. Phys.*, vol. 77, p. 056503, 2014.

[7] M. Tinkham, *Introduction to Superconductivity*, 2nd ed. New York: McGraw-Hill, 1996.

[8] Y. Hochberg, T. Lin, and K. M. Zurek, "Detecting sub-GeV dark matter with superconducting nanowires," *Phys. Rev. D*, vol. 94, p. 015019, 2016.

[9] M. W. Doherty *et al.*, "Temperature shifts of the resonances of the NV^- center in diamond," *Phys. Rev. B*, vol. 90, p. 041201, 2014.

[10] M. C. Cambria *et al.*, "Temperature-dependent phonon-induced relaxation of the nitrogen-vacancy center in diamond," *Phys. Rev. Lett.*, vol. 130, p. 256903, 2023.

[11] A. Jarmola *et al.*, "Temperature- and magnetic-field-dependent longitudinal spin relaxation in nitrogen-vacancy ensembles in diamond," *Phys. Rev. Lett.*, vol. 108, p. 197601, 2012.

[12] S. Ernst *et al.*, "Temperature dependence of photoluminescence intensity and spin contrast in nitrogen-vacancy centers," *Phys. Rev. Lett.*, vol. 131, p. 086903, 2023.

[13] J. Happacher *et al.*, "Temperature dependent photophysics of single NV centers in diamond," *Phys. Rev. Lett.*, vol. 131, p. 086904, 2023.

[14] L. Robledo *et al.*, "Spin dynamics in the optical cycle of single nitrogen-vacancy centres in diamond," *New J. Phys.*, vol. 13, p. 025013, 2011.

[15] M. P. Collings *et al.*, "A laboratory survey of the thermal desorption of astrophysically relevant molecules," *Mon. Not. R. Astron. Soc.*, vol. 354, pp. 1133–1140, 2004.